Aaron Glenn Goodman

curriculum vitae

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EDUCATION

MAGIST, University of California, Los Angeles 2025
(Master of Applied Geospatial Information Systems and Technologies)
Capstone (GEOG 498/499) – applied research and cartographic production for the Tahoe Regional Planning Agency (TRPA)

Graduate Certificate in Digital Humanities, University of California, Los Angeles 2025
Final paper, DH 250 Special Topics: Critical Cartography (Dr. Nour Joudah): *'Cartographic' Documents, Simulacrum, and Representation-Reality Feedback* – links early commercial recording expeditions with geographic knowledge production; mapping recording locations and personnel birthplaces from the Discography of American Historical Recordings

GIS and Geospatial Technology Certificate, UCLA Extension 2023

B.A., Ethnomusicology, University of California, Los Angeles 2021
Senior thesis (Dr. Timothy D. Taylor): *Music Festival Participation and Memory as a Work of Art: Embodying Community Encounters and Problematizing Application of the Carnivalesque* – reads music festival participation as embodied art of "play"

B.A., Anthropology, University of California, Los Angeles 2021

PROFESSIONAL EXPERIENCE

UCSB Library, Special Research Collections Santa Barbara, CA
Audio Digitization Technician (recurring grant-funded appointments) 12/2022 – 09/2023; 02/2024 – 03/2025; 11/2025 – 09/2026

Discography of American Historical Recordings (DAHR) / American Discography Project
- Controlled traffic of 78rpm discs from stacks to the Henri Temianka Audio Preservation Lab and back, keeping the discographic documentation and digitization phases of the workflow "in step" under National Endowment for the Humanities funding
- Transferred grooved media to archival digital audio
- Processed, organized, and quality-checked audio files, images, and metadata, then assembled them into Submission Information Packages before staging and publishing to the project repository; digitized 4,000+ unique recordings and 4,600+ label images in the first appointment
- Ingested externally created assets and metadata for post-custodial partnerships, e.g., the 2025 Dust-to-Digital release news.ucsb.edu/2025/022193
- Assessed and reprocessed content from older Submission Information Packages as project conventions evolved
- Rebuilt DAHR's public recording-locations map in ArcGIS Online; quality-checked the underlying place data and documented the geodata workflow for DAHR editors (see **Geospatial Portfolio**)

Archival Processing and Description
- Digitized and created item-level inventories for audiovisual materials in performing arts collections (see **Collections and Digital Projects**), including format identification and personnel/date research
- Applied controlled vocabularies and prepared descriptive metadata for EAD finding aids submitted to the Online Archive of California; crosswalked metadata between local and repository standards with FileMaker Pro and Python/ODBC scripting
- Staged and published digitized audio and metadata to public access platforms (Aviary) and internal repositories following UCSB Library SRC metadata conventions

Digital Asset Management and Systems
- Developed Python microservices with FFmpeg and ODBC to batch-process assets and metadata gitlab.com/ucsb-lib/src-avlab-microservices
- Built, maintained, and extended bespoke FileMaker Pro solutions for digital asset management and metadata remediation, including a complete overhaul of the cataloging and digitization staff's solution for the UCSB Cylinder Audio Archive cylinders.library.ucsb.edu and a solution for the Sound Recording Trade Catalogs collection
- Logged workflow status across multi-year, grant-cycle projects so that work was tracked and repeatable

Training, Supervision, and Documentation
- Trained and supervised student assistants (13 trained or co-trained) in the Henri Temianka Audio Preservation Lab, conservation lab, and stacks, including disc transfer and photography of disc labels and runoff areas
- Wrote and refined documentation for the physical handling and digitization of grooved media (wax cylinders, shellac, vinyl, lacquer) and magnetic media (open reel, audiocassette, VHS), and for the batch-processing and ingest (DAM) workflows; this documentation doubled as the student training curriculum

Ganna Walska Lotusland Montecito, CA
Library and Archival Collections Volunteer *05/2026 – 08/2026*
- Completed a folder-level inventory of Lotusland's non-living (cultural and historical) collections in CatalogIt

GEOSPATIAL PORTFOLIO

Full gallery with maps, reports, web apps, and notebooks at aggis.org/maps and aggis.org/files/notebooks

Projects

DAHR Recording Locations – ArcGIS Online Instant App, UCSB Library SRC 2026

- Rebuilt DAHR's public recording-locations map ahead of the deprecation of ArcGIS Web AppBuilder; quality-checked the place data behind nearly 1,000 recording locations and documented the geodata workflow so DAHR editors can maintain it adp.library.ucsb.edu/places, aggis.org/maps/dahr/2

Tahoe Regional Planning Agency capstone – GEOG 498/499, UCLA MAGIST 2025

- Transportation equity study; conifer encroachment analysis; lake water clarity with the Normalized Difference Turbidity Index
- Vikingsholm accessibility check: ArcGIS StoryMap proposing expanded parking at Emerald Bay State Park aggis.org/maps/vikingsholm
- Sea-level-rise animation and inundation risk assessment; operations dashboard; capstone video
- TRPA Data Explorer notebook pulling parcels, transit stops, and trail layers from TRPA ArcGIS REST services (Python, geopandas)
- Cartographic exercises in Adobe Illustrator: USGS-style topographic map, Tahoe land cover map, TART Mainline bus route map

Washington Land Cover Classification – random forest in Google Earth Engine, GEOG 415, UCLA MAGIST 2025

- Class-wide collaborative project: each student digitized seasonal ground truth for one Landsat path/row and year, then the class pooled its collections into a statewide training set spanning 13 WRS-2 footprints and 2014–2024
- Digitized 600 ground-truth points in six land cover classes across 13 cloud-free Landsat 8/9 scenes (four seasons, 2024); consolidated the class's 19 contributed collections by inspecting schemas, converting polygons to centroids, repairing property names and types, excluding collections without scene metadata, and screening coordinate errors
- Joined each point to its source scene, masked clouds with the QA_PIXEL band, scaled to surface reflectance, and sampled six bands plus Tasseled Cap components; ran a 30–150 m scale-sensitivity test and tuned a random forest classifier within Earth Engine memory limits (68% overall accuracy; confusion matrix, producer's/consumer's accuracy, kappa)
- Applied the model to 2017 and 2022 summer composites for post-classification change detection aggis.org/maps/lulcml, github.com/leependu/washington-lulc-ml-landsat

Istat Migration History Timeline – DH 201, UCLA Digital Humanities 2024

- Exploratory data visualization of Italian National Institute of Statistics (Istat) time series (1869–2014), published as an ArcGIS Online time-slider app aggis.org/maps/istat

DAHR Global Recording Explorer – ArcGIS Online web app 2023

- Joined DAHR's matrix table to a georeferenced place table to map recording locations for the nearly 70,000 78rpm records then digitized, with interactive widgets for label, date, and artist exploration; cleaned character-encoding and timestamp edge cases aggis.org/maps/dahr

Topics

Cartographic design and historical maps – GEOG XL 180/181, UCLA Extension 2022 – 2023

- Georeferenced Sanborn fire insurance maps for Redlands, CA (QGIS)
- Olympic National Park unofficial visitor's map (Adobe Illustrator); personal projects: Strava-derived bicycle heat maps from harvested GPX routes aggis.org/maps/bike; speculative sea-level-rise world-building maps aggis.org/maps/rpg

Geospatial statistics with R – GEOG 413, UCLA MAGIST 2025

- Kernel density estimation, spatial autocorrelation (Moran's I / LISA), and spatial regression reports (RStudio)

Remote sensing – GEOG 411, UCLA MAGIST 2024

- Supervised object-based classification with Landsat 8; vegetation (NDVI) change detection with NAIP imagery (ArcGIS Pro)

Python geospatial tools – GEOG 412, UCLA MAGIST 2024

- Jupyter notebooks: Airbnb exploration and local-area analytics tools (Los Angeles, Amsterdam); 2020 U.S. presidential election maps; North American Ski Explorer with geocoded radius search; World Bank API mapping tool (geopandas, folium, requests)

Interactive web maps – GEOG 414, UCLA MAGIST 2024

- Leaflet: 2020 Moldova presidential runoff results; socioeconomic marginalization in Mexico with user-selectable variables and color ramps aggis.org/maps/unit2a, unit2b
- D3: evolution of income and income inequality, 1800–2010 (Gapminder); time series of Italian migration, 1900–2014 (Istat) aggis.org/maps/unit4, unit5
- Hosted WMS/WFS services with GeoServer and PostGIS; deployed a web mapping stack on AWS EC2 (nginx, Tomcat, Let's Encrypt)

Geospatial database design – GEOG 410, UCLA MAGIST 2023

- Designed and documented a PostgreSQL/PostGIS geospatial database; spatial SQL, views, raster analysis, indexing, security

Advanced and intermediate GIS – GEOG XL 181A/181B, UCLA Extension 2022 – 2023

- Digitized warehouse structures in San Bernardino, CA (final project); Moran's I / LISA report for the 2016 U.S. presidential election (GeoDa); fire hazard mapping for residential parcels in Riverside, CA; site suitability analysis for outdoor recreation; viewshed analysis of T-Mobile cellular coverage; interpolation with IDW and empirical Bayesian kriging; network analysis for retail catchment areas; cost-path analysis across the Darién Gap; environmental health atlas with map series

INSTRUCTION, WORKSHOPS, AND PRESENTATIONS

Instruction and Training

- Trained or co-trained 13 student assistants in grooved-media handling, disc transfer, label photography, and batch-processing and ingest workflows; wrote the handling and workflow documentation used as training material – UCSB Library SRC, 2022 – 2026
- Documented the place-data quality-check workflow behind DAHR's recording-locations map so project editors can maintain the geodata – UCSB Library SRC, 2026
- Designed and delivered a beginner workshop on command-line media tools for learners with no prior experience: *yt-dlp and FFmpeg: Harvesting CC-BY Licensed Video from YouTube for Documentary Purposes* – DH 250 Special Topics: Professional Trajectories, UCLA, 2024

Presentations

- MAGIST capstone video presentation on applied GIS for the Tahoe Regional Planning Agency – GEOG 499, UCLA, 2025
- *'Cartographic' Documents, Simulacrum, and Representation-Reality Feedback* – DH 250 Special Topics: Critical Cartography, UCLA, 2025
- *Italian Migration and the Life of the Italian Town* – with C. Smith and D. Sparks; DH 201 Core Seminar in Digital Humanities, UCLA, 2024

TECHNICAL SKILLS

Archives and Collections: FileMaker Pro, CatalogIt, OpenRefine, Aviary; EAD/OAC finding aids; controlled vocabularies; metadata crosswalking and remediation
Audio and Video: FFmpeg, Steinberg WaveLab & Cubase, Logic Pro, Audacity, Renoise; recording, mixing, and mastering; video editing and stop-motion animation
Code Languages: Python, SQL, JavaScript, HTML/CSS, R, LaTeX; Unix and PowerShell shells; Git
Specialties: scripting and batch processing, data parsing/transformation/visualization, REST APIs and ODBC, machine learning (random forest classification)
Desktop GIS: ArcGIS Pro, QGIS; porting to Adobe Illustrator for refined cartography or animation
Geospatial Analysis: PostgreSQL/PostGIS, GeoDa, RStudio, Google Earth Engine, geopandas, rasterio
Web Mapping: ArcGIS Online (StoryMaps, Dashboards, Web AppBuilder [legacy]), Leaflet, D3, GeoServer, folium
Adobe CC and Office: Illustrator, Photoshop, Premiere, Dreamweaver, Bridge; Microsoft Office, Google Workspace, LibreOffice

COLLECTIONS AND DIGITAL PROJECTS

Discography of American Historical Recordings (American Discography Project), UCSB Library SRC 2022 – 2026
- Online 78rpm discography with 98,000+ recordings digitized and streaming; physical preservation, audio digitization, traffic control, content management, post-custodial ingest adp.library.ucsb.edu

UCSB Cylinder Audio Archive, UCSB Library SRC
- Online wax cylinder database with 15,000+ recordings digitized and streaming; FileMaker Pro solution redesign, content management, metadata remediation cylinders.library.ucsb.edu

John Steinbeck SoundScriber Discs (Mss 164), UCSB Library SRC
- Dictation discs recorded by John Steinbeck in the 1940s; item-level inventory, audio digitization OAC finding aid, Aviary

Leon and Googie René Recordings (PA Mss 248), UCSB Library SRC
- Lacquer and vinyl records created by R&B artists Leon and Googie René and their contemporaries, 1924–1993; item-level inventory, format and personnel research OAC finding aid

Eula Beal Recordings (PA Mss 262), UCSB Library SRC
- Lacquer records created in the 1940s by contralto Eula Beal; item-level inventory OAC finding aid

Dame Judith Anderson Papers, Series XII: Audio Recordings (PA Mss 6), UCSB Library SRC
- 7-inch open reel tapes of stage performances by Dame Judith Anderson; item-level inventory, audio digitization OAC finding aid

AUDIOVISUAL PRESERVATION EXPERTISE

- **Grooved media:** transfer of pre-war and 78rpm-era recordings; evolution of recording and playback equalization curves after electrical recording arrived in 1925; coexistence of lateral- and vertical-cut discs before the industry settled on lateral cut; physical preservation and cleaning of shellac, wax cylinder, vinyl, and lacquer/acetate discs
- **Magnetic media:** transfer of audiocassette, open reel, and VHS; adoption timeline of common formats; climate-controlled storage and baking for sticky-shed syndrome
- **Post-processing and packaging:** FFmpeg and archive-grade DAWs and video editors; Digital Asset Management; Submission Information Package assembly; generation of preservation and access derivatives; local metadata conventions across several collections
- **Description:** item-level inventories; controlled vocabularies; EAD finding aids for the Online Archive of California; metadata crosswalking and remediation; workflow documentation for student training

COURSEWORK

- **UCLA, MAGIST (2023 – 2025):** Applied Spatial Data Science; Geospatial Databases and Data Management (GEOG 410); Programming for Spatial Data Science I (GEOG 412) and II (GEOG 414); Geospatial Imagery Analysis (GEOG 411); Applied Spatial Statistics (GEOG 413); Spatial Data Science Futures (GEOG 415, Google Earth Engine); Capstone I: Geospatial Research Methods (GEOG 498); Capstone II: Geospatial Capstone Project (GEOG 499, TRPA)
- **UCLA, Graduate Certificate in Digital Humanities (2024 – 2025):** Core Seminar in Digital Humanities; Professional Trajectories: Digital Humanities Methods in Practice; Introduction to Critical Cartography and Theories of Mapping
- **UCLA Extension, GIS and Geospatial Technology Certificate (2022 – 2023):** Introduction to GIS; Cartography; Intermediate GIS; Geospatial Databases and Enterprise GIS; Advanced GIS; Web Design

FUNDING AND AWARDS

- MAGIST True Bruin Scholarship ($3,848), University of California, Los Angeles, 2023

RESEARCH INTERESTS

- History of recorded music and the development of genre-specific ethos and aesthetics
- Formal, musical, and lyrical analysis of historical recordings cross-referenced with contemporaneous print and artistic materials
- Geography of the early recording industry: studios, broadcast stations, venues, and vendors; spatial exploration of discographic metadata
- Time-series and historical GIS for demographic and cultural change; critical cartography and data visualization pedagogy
- History of reference mapping, GPS, and navigation technology for everyday use

LANGUAGES

English (native); Spanish (conversational; heritage exposure at home plus three years of high school and university study)